PES UNIVERSITY

SOFTWARE ENGINEERING

LAB 7: FUZZING FOR INPUT VALIDATION BUGS

Lab screenshots

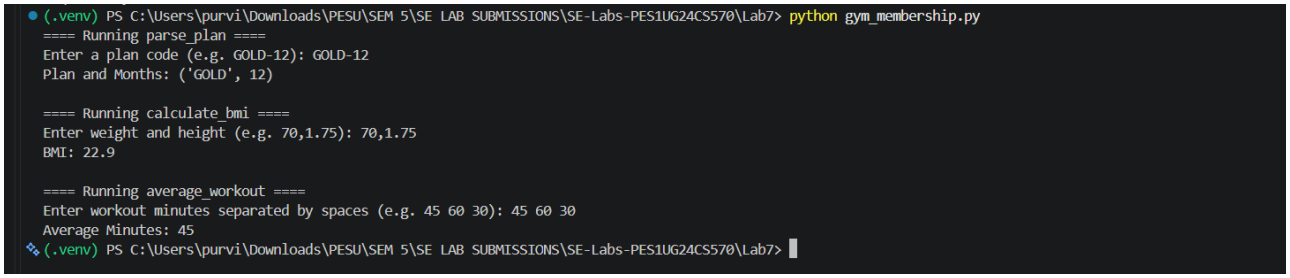
Name	G Purvi
SRN	PES1UG24CS570
Section	J
Course	Software Engineering UE24CS341A
Experiment	Lab 7 – Fuzzing for Input Validation Bugs

Screenshots

Screenshot 1: Original application output

File: before_fuzz_testing.png

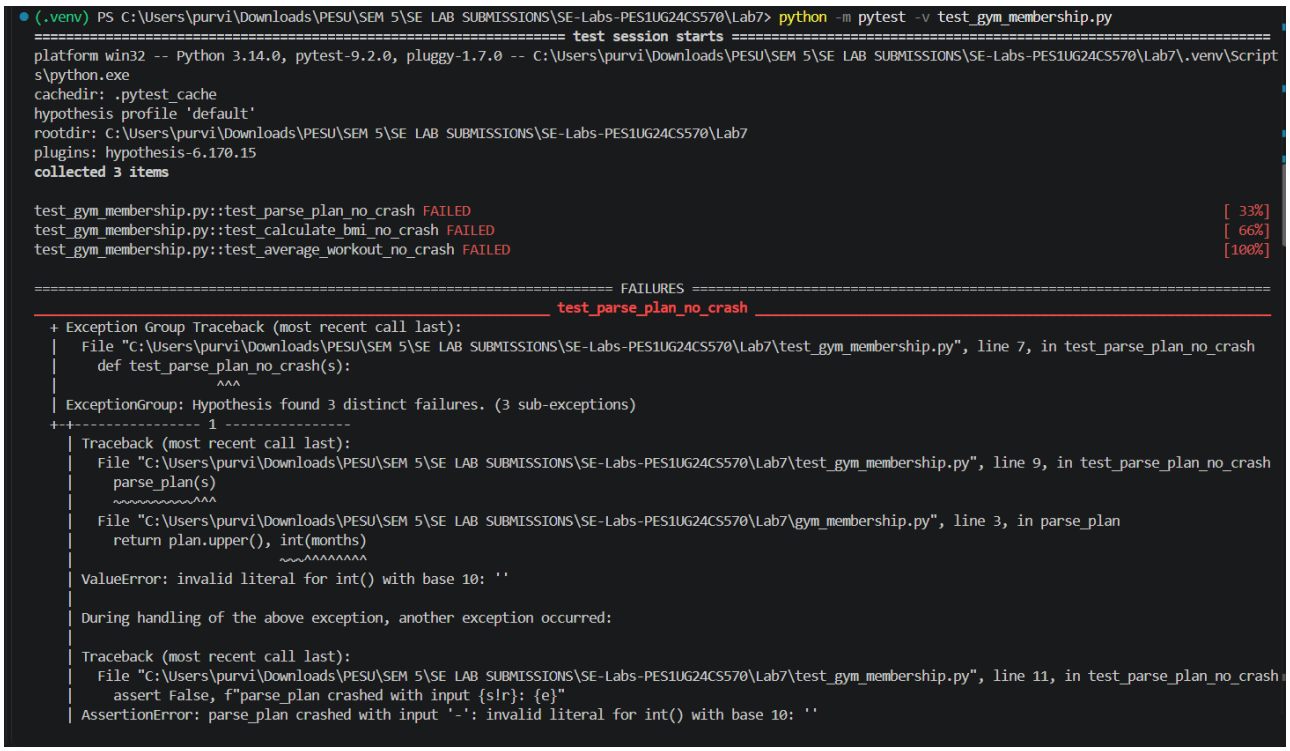
Evidence of the original application running successfully with valid sample inputs.



Screenshot 2: Initial fuzz-testing failures

File: fuzz_testing_failures1.png fuzz_testing_failures2.png

Evidence of the failures found by Hypothesis before input validation was added.



```
Problems  Output  Debug Console  Terminal  Ports

(.venv) PS C:\Users\purvi\Downloads\PESU\SEM 5\SE LAB SUBMISSIONS\SE-Labs-PES1UG24CS570\Lab7> python -m pytest -v test_gym_membership.py
    | Failing test case: test_average_workout_no_crash(
    |     s='',
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         C:\Users\purvi\Downloads\PESU\SEM 5\SE LAB SUBMISSIONS\SE-Labs-PES1UG24CS570\Lab7\test_gym_membership.py:26
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\purvi\Downloads\PESU\SEM 5\SE LAB SUBMISSIONS\SE-Labs-PES1UG24CS570\Lab7\test_gym_membership.py", line 25, in test_average_workout_no_
crash
    |     average_workout(s)
    |     ~~~~~~~~~~~~~~~~^^^
    |   File "C:\Users\purvi\Downloads\PESU\SEM 5\SE LAB SUBMISSIONS\SE-Labs-PES1UG24CS570\Lab7\gym_membership.py", line 10, in average_workout
    |     values = [int(m) for m in minutes.split()]
    |                              ^^^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\purvi\Downloads\PESU\SEM 5\SE LAB SUBMISSIONS\SE-Labs-PES1UG24CS570\Lab7\test_gym_membership.py", line 27, in test_average_workout_no_
crash
    |     assert False, f"average_workout crashed with input {s!r}: {e}"
    | AssertionError: average_workout crashed with input None: 'NoneType' object has no attribute 'split'
    | assert False
    | Failing test case: test_average_workout_no_crash(
    |     s=None,  # or any other generated value
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         C:\Users\purvi\Downloads\PESU\SEM 5\SE LAB SUBMISSIONS\SE-Labs-PES1UG24CS570\Lab7\test_gym_membership.py:26
    +----------------------------------
=========================== short test summary info ============================
FAILED test_gym_membership.py::test_parse_plan_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
FAILED test_gym_membership.py::test_calculate_bmi_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
FAILED test_gym_membership.py::test_average_workout_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
============================== 3 failed in 5.83s ===============================
(.venv) PS C:\Users\purvi\Downloads\PESU\SEM 5\SE LAB SUBMISSIONS\SE-Labs-PES1UG24CS570\Lab7>
```

Screenshot 3: Successful fuzz-testing run

File: after_fuzz_testing.png

Evidence showing all three fuzz tests passing after the application was corrected.

```
(.venv) PS C:\Users\purvi\Downloads\PESU\SEM 5\SE LAB SUBMISSIONS\SE-Labs-PES1UG24CS570\Lab7> python -m pytest -v test_gym_membership.py
============================= test session starts ==============================
platform win32 -- Python 3.14.0, pytest-9.2.0, pluggy-1.7.0 -- C:\Users\purvi\Downloads\PESU\SEM 5\SE LAB SUBMISSIONS\SE-Labs-PES1UG24CS570\Lab7\.venv\Script
s\python.exe
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: C:\Users\purvi\Downloads\PESU\SEM 5\SE LAB SUBMISSIONS\SE-Labs-PES1UG24CS570\Lab7
plugins: hypothesis-6.170.15
collected 3 items

test_gym_membership.py::test_parse_plan_no_crash PASSED                  [ 33%]
test_gym_membership.py::test_calculate_bmi_no_crash PASSED               [ 66%]
test_gym_membership.py::test_average_workout_no_crash PASSED             [100%]

============================== 3 passed in 0.81s ===============================
(.venv) PS C:\Users\purvi\Downloads\PESU\SEM 5\SE LAB SUBMISSIONS\SE-Labs-PES1UG24CS570\Lab7>
```

Screenshot 4: Corrected application output

File: after_fuzz_fix.png

Evidence that the corrected application still works with valid sample inputs.

```
(.venv) PS C:\Users\purvi\Downloads\PESU\SEM 5\SE LAB SUBMISSIONS\SE-Labs-PES1UG24CS570\Lab7> python gym_membership.py
==== Running parse_plan ====
Enter a plan code (e.g. GOLD-12): GOLD-12
Plan and Months: ('GOLD', 12)

==== Running calculate_bmi ====
Enter weight and height (e.g. 70,1.75): 70,1.75
BMI: 22.9

==== Running average_workout ====
Enter workout minutes separated by spaces (e.g. 45 60 30): 45 60 30
Average Minutes: 45
(.venv) PS C:\Users\purvi\Downloads\PESU\SEM 5\SE LAB SUBMISSIONS\SE-Labs-PES1UG24CS570\Lab7>
```